

ISHAN AGARWAL

Singapore | ishan_agarwal@u.nus.edu

ishan-agarwal.com | linkedin.com/in/ishan-agarwal-nus | github.com/ishan-agarwal-05

Final-year NUS Computer Science student graduating May 2027, with four software internships across robotics simulation, backend and applied AI, including six months building digital twin tooling at Hyundai's R&D centre.

EDUCATION

National University of Singapore

Aug 2023 – May 2027 (Expected)

Bachelor of Computing (Honours), Computer Science; Minors in Mathematics and Quantitative Finance Singapore

- Focus areas: Artificial Intelligence, Computer Security. Coursework: Data Structures & Algorithms, Operating Systems, Networks, Information Security, Machine Learning, Natural Language Processing, Cryptography.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, SQL, Apex, LaTeX

Backend, Web & Infrastructure: Flask, Node.js, React, Flutter, REST APIs, Docker, Git, CI/CD, MySQL

AI, Data & Simulation: PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, LangChain, LangSmith, NumPy, Pandas, NVIDIA Isaac Sim, Omniverse Kit, USD

EXPERIENCE

Hyundai Motor Group Innovation Center Singapore (HMGICS)

Jan 2026 – Jun 2026

Digital Twin & Simulation Engineering Intern | Python, NVIDIA Isaac Sim, OmniGraph, USD Singapore

- Built an AI code-generation workflow that turns Kawasaki AS robot programs into fully wired Isaac Sim control graphs, cutting **roughly one engineer-week of manual wiring per subprogram to about four hours**, review included. A single factory cell has 40+ subprograms.
- Scoped it alone from a one-line brief, compared **seven approaches**. An LLM agent writes a deterministic Python script per subprogram, steered by my model-agnostic harness of translation rules, node references and examples.
- Reverse-engineered **four undocumented OmniGraph and USD API behaviours** by diffing generated scene output against a hand-built baseline, then centralised the fixes into a shared module.
- Shipped the simulation reporting system and three Omniverse Kit UI extensions on a config-driven architecture every later feature reused. Reports generate per run with joint-angle charts, tool paths and a parts inventory.
- Audited asset traceability against an 80% target by cross-referencing **36 physical parts** against inventory systems, lifting coverage **from 31% to 44%** and showing the remaining gaps were missing data, not code defects.

TechFour Engineering Solutions

May 2025 – Jul 2025

Software Engineering Intern | Python, Flask, Flutter, MySQL India

- Built and deployed **three backend microservices** for user management (authentication, OTP, password reset), document handling and internal communications, built as reusable components so the company's later projects could drop them in instead of rewriting them.
- Integrated WhatsApp and email APIs and shipped the Flutter client.
- Added SonarQube quality gates and documented every endpoint in Swagger.
- Learned Linux, Nginx and load balancing in production from the engineer who ran the company's deployments.

PwC India

Dec 2024 – Jan 2025

AI Engineering Intern | Python, LangChain, LangSmith, Streamlit India

- Prototyped a retrieval-augmented generation assistant over **several hundred internal policy and research documents**, for policy lookup and for surfacing prior work so teams stopped repeating research. Built in LangChain, with LangSmith for tracing and offline evaluation.

PROJECTS

LectureAI | AI Lecture Notes Platform (Co-Founder)

Feb 2025 – Dec 2025

- Led full-stack development of a pipeline turning lecture recordings into study notes: FFmpeg audio, Whisper transcription, slide-aware transcript correction, then LLM summarisation, on FastAPI and React. Ran customer discovery across NUS cohorts; wound down in Dec 2025 as university integrations proved years away.

Extractive Question Answering | Margin-Triggered Reranking

Aug – Dec 2025

- Fine-tuned RoBERTa to **84.3 exact match** on SQuAD v1.1 and found the gold span already sat in the top five candidates for **95% of questions**. Built a bi-encoder reranker that fires only on low-margin predictions: it changed **55 of 10,570** answers (14 fixed, 1 broken) and left confident predictions untouched.

Fake News Detection | Metadata and Fairness Trade-offs

Jan – May 2025

- Benchmarked **18 model and feature setups** on the LIAR political-claims dataset, from TF-IDF to fine-tuned DistilBERT. Speaker metadata added **2 to 4 points** of macro-F1, partly by learning who is speaking, a fairness cost.

ADDITIONAL

NTSE Scholar, All India Rank 21 (National Talent Search Examination) | Salesforce Administrator & Developer certified
Cambridge English C2 Proficiency (Grade A) | Volunteer with at-risk youth, Teach SG